Dash Kamriani Beard

dashpk2@illinois.edu | linkedin.com/in/dpkb | (415) 622-8055

Education

University of Illinois Urbana-Champaign, B.S. in Computer Engineering, Minor in Astronomy — May 2025

Honors & Activities: *2023 Fiddler Innovation Fellowship Award, Formula SAE Team, ACM SIGArch*

Relevant Coursework: Computer Architecture, FPGA Lab, Operating Systems, Applied Machine Learning, GPU Programming (CUDA), Control Systems, Linear Algebra, Differential Equations, Algorithms

Work Experience

ASIC Design Engineer, Mill Computing — Aug 2025 – Dec 2025

- Implemented execution-stage operations and functional units for a novel low-power CPU using C++-based SystemVerilog code generation
- Wrote SystemVerilog testbenches to verify pipeline and execution-stage functionality
- Collaborated with compiler/simulator team to align hardware functionality with toolchain and ISA behavior

Contractor, Klaus Multiparking — Aug 2025 – Present

- Centralized customer contracts, payment records, and company metrics into a single internal system
- Redesigned an internal dashboard to help technicians and office staff coordinate work at scale
- Assisted in reverse-engineering Arduino-based parking controllers to reduce dependence on overseas parts
- Streamlined collections, contract renewals, technician dispatch, parts tracking, and service documentation, helping recover outstanding revenue and reduce service backlog

Research Intern, Lawrence Berkeley National Lab — Jun 2024 – Aug 2024

- Built a Jupyter → SLURM pipeline to batch-run fluid-flow carbon sequestration simulations on lab's HPC cluster
- Automated parameter sweeps for sensitivity analysis, eliminating manual setup
- Applied a Python genetic algorithm (GA) to determine CO_2 injection parameters for maximum sequestration
- Analyzed sensitivity and GA outcomes with mentors, presented findings at a technical forum in Pittsburgh

Lead Developer, Carle Illinois Medical School `tec.illinois.edu/news/55414` — Apr 2023 – Jun 2023

- Developed a companion app with React Native for an assistive anti-fatigue vest designed for surgeons
- Authored a suite of tools to collect, visualize, and analyze posture sensor data, facilitating vest interaction
- Structured a session system that lets multiple users track their posture data over time
- Device and app were presented at a forum in Chicago, won the Fiddler Innovation Fellowship Award

Projects

Out-of-Order RISC-V CPU `paper`

Designed out-of-order RISC-V processor with ERR architecture, implementing RV32IM spec.

- Implemented out-of-order pipeline with reservation stations, reorder buffer, and common data bus architecture
- Enhanced performance (48% IPC improvement, 57% delay reduction) with early branch recovery, 2-way superscalar backend, pipelined cache, and split load-store queue with dependency tracking
- Verified design using EEMBC Coremark, SystemVerilog constrained random TB, and custom assertion framework
- Synthesized at 588 MHz with multi-instruction commit capability and advanced memory disambiguation

Inverted Reaction-Wheel Pendulum `paper`

- Designed a feedback controller for a reaction-wheel pendulum using Lagrange-based modeling, state-space linearization, and Simulink
- Estimated friction and motor parameters from constant-speed experiments, then used polynomial fitting to recover smooth derivatives from noisy sensor data
- Implemented switching control for swing-up, reversing wheel rotation at peak motion and reducing stabilization time by 90%

Tetris on FPGA

Recreated Tetris as a MicroBlaze-based SoC on AMD Spartan-7 FPGA.

- Programmed AMD MicroBlaze processor IP in C to handle USB keyboard input via AXI Quad SPI
- Developed HDMI output pipeline using VGA-to-HDMI IP core with custom color mapping and pixel generation
- Integrated multiple AXI devices (timers, GPIO, UART, and interrupt controllers) for real-time gameplay

Technical Skills

General: SystemVerilog, UVM, RISC-V, C, C++, Python, CUDA, Linux Systems/Drivers, Arduino, OpenXR
Tools: Synopsys VCS, Verdi, and Design Compiler, Vivado, MATLAB, Simulink, Keras TensorFlow, git, GDB